Deepraj Pandey

deepraj.pandey_ug20@ashoka.edu.in | +91-732 708 0977

EDUCATION

ASHOKA UNIVERSITY
BSc. in Computer Science
May 2020 | Sonepat, India

D.A.V. PUBLIC SCHOOL
Grad. May 2017| Bhubaneswar, India

SKILLS

PROGRAMMING
Current:
Python • C • C++ • HTML • CSS • LaTeX
Familiar:
R • Shell • Javascipt • Android
Other:
Git • Processing • Google Apps Script
Arduino • Ruby on Rails • TI-BASIC
Systems/Simulators/Libraries:
MacOS • Linux • GCP • Gem5
NVMain • Champsim • JOS Kernel
Keras • Py Cryptography

COURSEWORK
Software Debugging*
Program Analysis*
Operating Systems
Intro Machine Learning
Computer Security
Algorithm Design
Advanced Programming
Programming Language Design

** Adv elective in first year.*

VOLUNTEERING
Banjaara - Ashoka Fest, 2018, 2019
Alumni Outreach Team ('18), Head of Tech Team ('19)

MLH Local Hack Day | December, 2017
Organised a 12 hour internal hackathon on campus in first semester.

Ashoka Sports Fest | October, 2017
Handled logistics for Agneepath - annual inter-University sports event.

Rotary IWC | Bhubaneswar
400+ hours of volunteering 2014 - 2017

Quizmaster | High School
Organised the flagship annual tech quiz from 2014-2017

MISCELLANEOUS
Air Rifle Shooting
Hacktoberfest - '17, '18, '19
School Football Team (Captain)
INCA Map Quiz '15 - National Rank 1

EXPERIENCE

ALUMNI RELATIONS COMMITTEE | Core Member
September 2019 - Present | Ashoka University, India
- 8 member team of undergraduates and fellows of Ashoka reporting to the Alumni Relations Office responsible for Alumni-student interactions.
- Working with 2100+ alumni for events/mentorships on/off campus.
- Designed the algorithm and set up system to automate alumni-student matching for the annual mentorship program.

DEPARTMENT OF COMPUTER SCIENCE | Research Assistant
May 2019 - Aug 2019 | Ashoka University, India
- Worked with Prof. Manu Awasthi on non-volatile memory architecture in hand-held devices.
- Designed an interface to gem5 that sets up arbitrarily deep cache hierarchies along with NV memory (via NVMain).
- Code reviewed and in works to be a part of META, a memory simulation tool.

OFFICE OF ACADEMIC AFFAIRS | Internship
February 2019 - April 2019 | Ashoka University, India
- Worked with Prof. Hemanshu Kumar and the Office to design an algorithm and set up a system for faculty advisors and the OAA to check course credit requirements of final year students for graduation.
- Code submitted and reviewed. Later taken up by IT after end of internship.

MYTRAH MOBILITY | Summer Intern
May 2018 – August 2018 | Gurgaon, India
- Worked on the Electric Mobility company's fleet electrification analysis tool.
- Parsed and prepared data on 4000 public buses in multiple Indian cities. Code reviewed and pushed to production.
- Led a team of all interns to design a long-term plan for non-profit wing, EMA.
- Planned the company's flagship hackathon before end of internship.

STUDENT AMBASSADORS PROGRAMME April 2018 - August 2019 |
Ashoka University, India
- Selected as a student ambassador under the University Outreach Team.
- Engaged with prospective students and parents on diverse subjects about student life at Ashoka.
- Assisted Admissions team by mentoring the incoming cohort in the months before their first semester.
- Represented the university to various stakeholders through interactions and campus tours.

LEADERSHIP

WOMEN IN COMPUTING SOCIETY | Head of Events
August 2019 - Present | Ashoka University, India
- Previously co-head, currently heading a team of 8 responsible for planning and organising workshops, guest talks, and socials, usually around promoting women representation in Tech.
- Have personally organised workshops on Git, Programming, and on Web Development since 2017 on behalf of the society.

HULT PRIZE ASHOKA | Outreach
September 2019 - December 2019 | Ashoka University, India
- Organised the OnCampus Event for Hult Prize - an international social entrepreneurship competition.
- Engaged with entrepreneurs & business leaders and built a panel of judges for the event.

PROJECTS

CNN - Chars74K | Colab
81.4% val_accuracy
Learning about CNN by experimenting with the architecture and data.

Garbage Collector | GitHub
Implemented the mark and sweep phases of a stop the world garbage collector for Programming Language Design and Implementation class.

Gina | GitHub
A bot built on first principles for syntax analysis and to learn a new language based on user translations.

Data Encryption Standard Algorithm | GitHub
A DES implementation in C++ based on basic function template for a pset.

Network Design on Cisco Packet Tracer
Designed multiple networks on Cisco Packet Tracer - internetworks with DHCP servers, web hosting, DNS servers, VLAN, NAT enabled routers (private networks) etc.

OS Labs on xv6
Coursework similar to MIT 6.828. Edited kernel code to add physical page allocators, , interrupt handlers, system calls, page faults, and context switches.

Trace Driven Cache Simulator | GitHub
Simple, proof of work trace based cache simulator with 32KB L1 caches (4 words).

Music Society Website
Built the music society's website and set up Google Apps Script to automatically fill slots on public facing spreadsheet on the site every time someone books a music room slot through a google form.

Biology Society Blog
Built the biology society's blog using Jekyll with templating for easy maintenance and post additions. Hosted on GitHub pages.

COMMUNITY

Computer Architecture Reading Group - Summer '19, Spring '20

Google Developer Group, Bhubaneswar: 2015 - Present

Google Develper Day: 2016 (Bhubaneswar), 2019 (Gurgaon)

LINKS

Github:// DeeprajPandey
LinkedIn:// DeeprajPandey
Academia:// DeeprajPandey
Twitter:// @DeeprajPandey

BANJAARA - ANNUAL FEST | HEAD OF TECH TEAM
October 2018 – March 2019 | Ashoka University, India

- Led a team of 6 developers, designers for the annual fest.
- Drove the development of the fest website, a full stack registration portal (Laravel, Heroku, Google Cloud Storage), and hardware projects for fest day (arduino powered photo booth with a pull switch capture setup).

HACKDAV | FOUNDER
October 2015 – Present | Bhubaneswar, India

- Founded the state's first hackathon for high school students.
- Applicants from across India and 5 other countries.
- Led the team that raised **INR 1,800,000** in sponsorships for 2017.
- Major sponsors include DigitalOcean, Soylent, JetBrains, MLH, Balsamiq, Hackerearth along with Google and Microsoft Developer Groups.

CONFERENCES

INTERNATIONAL SYMPOSIUM FOR COMPUTER ARCHITECTURE | UNDERGRADUATE WORKSHOP ATTENDEE
June 2019 | Phoenix, USA

- One of the 42 undergrads selected from across the world for the **first undergraduate architecture workshop** at ACM FCRC - **ISCA 2019**.
- Awarded a substantial scholarship by UC Davis + travel grant by Ashoka for attending.

MLH HACKCON VI | GITHUB SCHOLAR
June 2018 | Pennsylvania, USA

- Awarded the GitHub Scholarship for attending the student hackathon organiser conference organised by MLH and GitHub.
- Only student organiser representing a high school hackathon (HackDAV).
- Ideated a hacker exchange program with UMass & UTexas with GitHub.

ISTEM HACKATHON | WINNER
Jan 2019 | IIIT Bangalore, India

- Worked in a team with partially blind participants to build an accessibility web app that extracts text data from coding tutorial videos for viewers with visual impairments.
- Teams with partially blind programmers and incorporated their insights in the product. Won the hackathon which had 75+ participants including Graduate students and professionals from across the country.

NOTRE DAME LEADERSHIP SEMINARS | PARTICIPANT
July 2016 | Notre Dame, USA

- Only student from India (among the 98 from across the world) selected for a **fully-funded** summer program at the University of Notre Dame.
- Seminar on the ethical concerns of space exploration under **Prof. Don Howard** and **Dr. Jessica Baron**.

PROJECTS

TOPIC CLUSTERING ON ENRON DATASET | GITHUB LINK
November 2019

- Worked in a team of 3 on the topic modelling problem for Intro to ML.
- Tried different approaches and decided on using Latent Dirichlet Allocation which gave the best results.
- Links to **Source Code**, **Report**, **clustering visualisation**.

DISTRIBUTED SERVER | GITHUB LINK
November 2019

- Built a fully distributed, multithreaded server communicating over TCP.
- Servers run simultaneously on multiple devices and handle multiple clients.
- Has a request admission phase that accepts client-server communication and a time-stamp based consensus phase where servers get the same copy of the data.